Ridgeline Weather Network Health Briefing

Spring Climate Summary Decisions & Volunteer Technician Dispatch Plan

Coordinator Action Brief
6-Week Audit: Mar 01 – Apr 11, 2025
Prepared for Monday Meeting

✔ Decision 1: Spring Climate Summary

88.4% of network data is publication-grade. Saddle Ridge (S01) is 100% sound. Three stations (S02, S06, S07) are fully rescued via mathematical formulas. Corrupted windows at S03 (frozen temp), S04 (clogged rain), and S05 (waterlogged humidity) are isolated and excluded.

➔ Decision 2: Technician Dispatch

Dispatch technician to S03 (Valley Library). S03 has an active, recurrent temperature sensor disconnect (-40°C/+85°C spikes and 80-hour freeze). All other station errors are solved in software; S03 cannot and requires physical servicing.

Station-by-Station Quality Assessment & Inclusion Rules

Station	Elevation / Role	Summary Status	What Happened	Action for Spring Climate Summary
S01 Saddle Ridge	2913 m • Alpine Hut	INCLUDE (100%)	Extreme cold (-18.7°C) and low pressure (~709 hPa) look unusual but match alpine physics.	**Include all data without changes.** Confirmed genuine alpine climate record.
S02 North Mesa	1623 m • School	CORRECT PRESS.	Swapped uncalibrated pressure board Mar 21 at 11am local (+6.66 hPa jump).	**Subtract 6.66 hPa** from pressure after Mar 21 17:20 UTC. Other channels 100% good.
S03 Valley Library	1523 m • Valley Ref.	TRIM TEMP	Digital probe glitched (-40°C, +85°C) and froze at 6.5°C for 80 hrs (Mar 18–21).	**Exclude Mar 18 06:00 – Mar 21 14:00 temp** and 4 glitch spikes. Rest 100% good.
S04 Community Gdn	1763 m • Foothills	EXCLUDE RAIN	Funnel clogged with leaves/nest until cleaned Apr 2 (33.4 mm surge). Fence repainted.	**Exclude rainfall prior to Apr 2 16:30.** Note subtle daytime fence heating.
S05 Eastside Rec	1644 m • Plains Edge	TRIM RH + META	Install sheet had lat typo (39.57 vs 39.754). Humidity saturated >100% after Mar 7 storm.	**Exclude RH from Mar 7 12:00 to Mar 11 00:00.** Update latitude to 39.7540° N.
S06 Fire Station	1891 m • South Hills	SHIFT TIMESTAMPS	Rebooted to Mountain Time (UTC-6) after battery swap Mar 24; clock ran 6h behind.	**Add 6 hours to timestamps** from Mar 24 10:10 onwards. Recovers 100% of data.
S07 Riverbend Farm	1525 m • North Valley	SCALE WIND ×3.6	Replacement anemometer fitted Mar 26 outputs pulses in m/s rather than km/h (dropped 3.6×).	**Multiply wind speed by 3.6** from Mar 26 15:30 onwards. Completely restores record.
S08 Plains Grange	1437 m • East Plains	DROP DUPLICATE	35.8-hr modem blackout Mar 16–18 (SIM expired). Apr 05 data re-appended twice at end.	**Drop trailing duplicate block.** Mark Mar 16–18 missing. Remainder is 100% sound.

Permanently Exclude from Climate Means:
- S03 Temperature: 80-hour frozen flatline (Mar 18 06:00 – Mar 21 14:00) & 4 spikes
- S04 Precipitation: Entire rain record prior to Apr 02 16:30 (clogged funnel + surge)
- S05 Humidity: Saturated period (Mar 07 12:00 – Mar 11 00:00, values 100–105%)
- S08 Telemetry: Missing SIM blackout gap (Mar 16 13:50 – Mar 18 01:40) & duplicate day

Recovered in Full via Script Formulas:
- S06 Clocks: Timestamps shifted +6 hours → 100% complete time-series restored
- S07 Wind: Multiplied by 3.6 (m/s → km/h conversion) → matches regional baseline
- S02 Pressure: Subtracted 6.66 hPa → barometric elevation curve seamlessly aligned
- S05 Coordinates: Corrected install sheet latitude typo to 39.7540° N

Volunteer Technician Work Order & Dispatch Plan

Mission Plan: Station S03
Location: Valley Library
Estimated Time: 45–60 Minutes

Where to send our one technician next week • 1-Hour Step-by-Step Task List

Why S03 and NOT the Second-Best Choice?

Every other station's anomaly is **mathematically deterministic and already resolved**: S06 was a fixed clock shift (+6h), S07 was a known unit scale (3.6×), S02 was a static board offset (-6.66 hPa), and S04/S08 issues are already past.

S03 is the only station with an ACTIVE, RECURRING HARDWARE FAILURE. Its digital temperature probe suffered -40°C open-circuit disconnections and +85°C reset defaults on Mar 05, Mar 12, Apr 02, and Apr 08, plus an 80-hour total freeze. Software cannot invent missing weather data. If not physically repaired, S03 will continue losing spring data.

What to Check FIRST (Minute 0–10):
Inspect screw terminals inside the logger enclosure and radiation shield strain relief. A loose screw or corroded ground on a 1-wire digital bus produces intermittent disconnects (-40°C) and reset register states (+85°C).

1-Hour Field Protocol for the Volunteer

1. Min 00–10 (Terminal Inspection): Open enclosure, inspect probe wiring to terminal block. Check for loose strands, moisture ingress, or oxidized contacts. Gently tug each wire.
2. Min 10–25 (Sensor Replacement): Swap the Dallas DS18B20 temperature probe with the pre-tested spare from the kit. Tighten screw terminals firmly.
3. Min 25–35 (Radiation Shield Check): Check louvers of radiation shield for spider webs or debris that trap heat/moisture against probe. Clean with brush.
4. Min 35–50 (Real-Time Diagnostic): Power on logger display. Confirm live temperature updates every 10s and tracks ambient (compare with handheld reference).
5. Min 50–60 (Close & Sign-off): Reseal weatherproof grommet with silicone grease, verify logger status LED blinks green, log replacement in notebook.

Equipment to bring: Small flathead screwdriver, spare DS18B20 probe, handheld reference thermometer, lint-free cloth, desiccant pouch.

Secondary Backlog & Remote Resolutions (No Field Visit Required)

Station	Reported Issue	Why Remote Fix is Better than a Field Visit	Action Taken in Data Pipeline
S06 Fire Station	Clock 6h behind UTC	The data is completely sound; a site visit would only take 5 mins to press buttons, but our script resolves all historical data cleanly.	Next routine battery change will re-sync clock; data corrected via automated +6 hr pipeline rule.
S07 Riverbend Farm	Anemometer reads low	New anemometer is mechanically sound and spinning freely; the only issue is the logger interpreting m/s pulses as km/h.	Scaled by factor of 3.6 in data ingestion. Next site visit can update logger firmware pulse constant.
S02 North Mesa	Pressure +6.66 hPa high	Uncalibrated spare board has a steady, pure linear offset. A site visit without a high-precision lab calibrator cannot fix it on-site.	Subtracted static 6.66 hPa offset in software. When calibration kit is available later this summer, calibrate properly.
S04 Community Gdn	Rain blockage & painted post	Funnel was already cleared by volunteer on Apr 2. Post repaint is permanent; daytime solar reflection is noted in metadata.	Excluded invalid historical rain; current rain gauge is working. Siting note added to annual report.

Guidance for the Coordinator at Monday's Meeting:
You can confidently reassure the school committee that our network data is sound and defensible. By applying these precise mathematical adjustments, we preserve over 95% of the spring weather history without sweeping flawed data under the rug. Sending our volunteer to S03 protects our upcoming summer dataset before severe heatwaves begin.